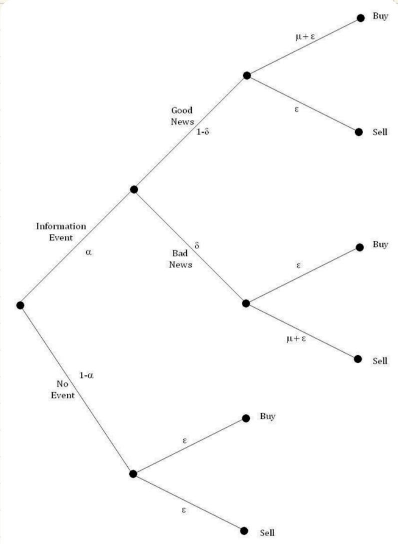
VPIN: Volume Sycnnised Probability of Informed Trading

⊛ PIN → VPIN



The sequential trading diagram of the 1996 PIN model

$$E[S_t] = (1-\alpha)S_0 + \alpha_t\left[\delta_t S_B + (1-\delta_t) S_G\right]$$

- Informed Traders,

  arrive $\sim$ Poisson(μ)

- Uninformed Traders:

  arrive $\sim$ Poisson(ε)

- Avoid losses from Informed traders

As Makers:

breakeven 点:

Bid: $E[B_t] = E[S_t] - \dfrac{\mu\alpha_t\delta_t}{\varepsilon + \mu\alpha_t\delta_t}\left(E[S_t] - S_B\right)$

Ask: $E[A_t] = E[S_t] + \dfrac{\mu\alpha_t(1-\delta_t)}{\varepsilon + \mu\alpha_t(1-\delta_t)}\left(S_G - E[S_t]\right)$

此时: Spread $= E[A_t - B_t]$

当 $\delta_t = \frac{1}{2}$: $= \dfrac{\alpha_t\mu}{\alpha_t\mu + 2\varepsilon}(S_G - S_B)$

$= \dfrac{1}{1 + 2\frac{\varepsilon}{\alpha_t\mu}}(S_G - S_B)$

当: $\alpha, \mu \uparrow$

更多信息事件与交易者

⇒ MM 扩大报价 Spread

⇒ Makers 的 Range:

$= \dfrac{\alpha_t\mu}{\alpha_t\mu + 2\varepsilon}$

→ Informed Traders 的期望到达率

↘ 总交易到达率 (Informed + Noise)

⊛ VPIN 使用 Tips:

1: 用 volume bars, 无需 tick

2: Bulk classification 而非 tick-rule

3. ln(VPIN) 才对 return 有预测力

✱ VPIN. a high-frequency estimation for PIN

PIN → VPIN: (作者认为: VPIN 是对 PIN 的 a good high-frequency estimation (摆脱了 MLE 估计参数步骤)

Informed Traders 通常单边交易

⇒ Order Imbalance 是知情交易的代理

⇒ 引入 Trading Classification 拆解 Volume

⇒ $\mathrm{VPIN} \propto \frac{\sum |V_\tau^B - V_\tau^S|}{nV}$ (按 Volume Bulks, n 是个超参数)

noise Traders 随机买卖

聚集净买入/卖出 ⇒ 大概率 Informed Traders

假设: Price 幅度 ~ 主买/卖意愿

High-Frequency Estimation

In the year 2010, David Easley, Marcos Lopez de Prado and Maureen O'Hara proposed a high-frequency estimate of **PIN**, which they denominated **VPIN**. This procedure adopts a *volume clock* which synchronizes the data sampling with the market activity, as captured by regular volume *buckets*.

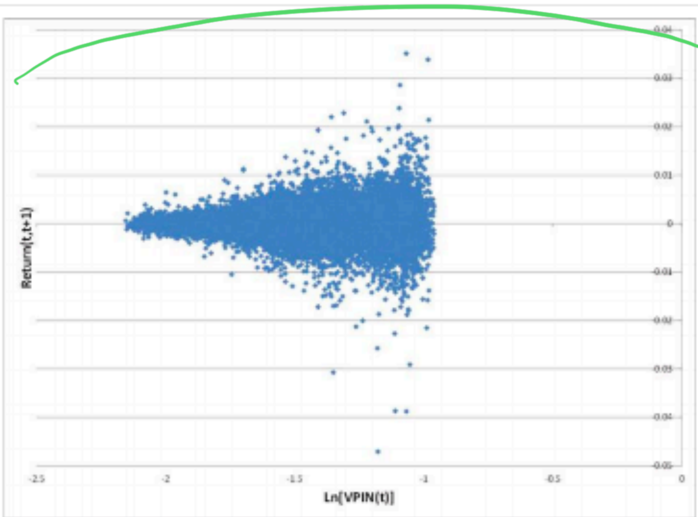
This is a form of subordinated stochastic process that departs from the standard *chronological clock* (i.e., sampling at regular time periods), and can be traced back to the work of Benoit Mandelbrot.

自然时间: 1 min K-bar

These authors begin by dividing a sample of *volume bars* (or similarly, time bars) in *volume buckets* (groups of trades such that each group contains the same amount of traded volume). Because all buckets are of the same size, V,

$$\frac{1}{n}\sum_{\tau=1}^{n}\left(V_\tau^B + V_\tau^S\right) = V = \alpha\mu + 2\epsilon\ .$$

where n is the number of volume bucket used to estimate **VPIN**. The procedure requires a method to split volume in *buys* and *sells*. Rather than using the *Tick-rule*, *Lee-Ready* or other trade classification techniques, they propose a new volume classification method called *Bulk Volume Classification*. This departs from standard trade classification schemes in two ways: First, volume is classified in bulk, and second this methodology classifies part of a bar's volume as *buy*, and the remainder as *sell*. Empirical studies have shown *Bulk Volume Classification* to be more accurate than the *Tick-rule*, despite of not requiring level-1 tick data (only bars).[3] Within a volume bucket, the amount of volume classified as buy is



Periods of high VPIN values (x-axis) are followed by increased volatility (y-axis)

问题:

1. 时间均匀, 但交易不均匀
2. 物理时间下:

Return ~ 尖峰厚尾

不服从正态分布假设

⇓

volume clock

⇓

设计 volume Buckets

e.g. Regular volume Buckets

⇓

由 volume bucket 计算

volume Bars

$$V_\tau^B = \sum_{i=t(\tau-1)+1}^{t(\tau)} V_i Z\left(\frac{S_i - S_{i-1}}{\sigma_{\Delta S}}\right)\ .$$

where $t(\tau)$ is the index of the last (volume or time) bar included in bucket τ, V_τ^B is the *buy volume* (traded against the Ask), V_i is the total volume per bucket, Z is the Standard normal distribution, and $\sigma_{\Delta S}$ is the standard deviation of price changes between (volume or time) bars. Because all buckets contain the same amount of volume V,

$$V_\tau^S = \sum_{i=t(\tau-1)+1}^{t_\tau} V_i\left(1 - Z\left(\frac{S_i - S_{i-1}}{\sigma_{\Delta S}}\right)\right) = V - V_\tau^B\ .$$

Since $\mathrm{E}[|V^S - V^B|] \approx \alpha\mu$, and $PIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon}$, it can be shown that VPIN is a good estimate of PIN, with

$$VPIN = \frac{\sum_{\tau=1}^{n}|V_\tau^S - V_\tau^B|}{nV}\ .$$

④

VPIN 到 Order Flow toxicity
Adverse Selection

the crash. According to this paper, *order flow toxicity* can be measured as the probability that informed traders (e.g., hedge funds) adversely select uninformed traders (e.g., Market makers). For that purpose, they developed the VPIN Flow Toxicity metric, which delivers a real-time estimate of the conditions under which liquidity is being provided. If the order flow becomes too toxic, market makers are forced out of the market. As they withdraw, liquidity disappears, which increases even more the concentration of toxic flow in the overall volume, which triggers a Feedback mechanism that forces even more market makers out. This cascading effect has caused hundreds of liquidity-induced crashes in past, the *flash crash* being one (major) example of it. One hour before the *flash crash*, order flow toxicity was at historically high levels relative to recent history.

Toxic 是一个逐渐累积的动态过程

⇒ Liquidity Crises

Although VPIN metric is conceived for the HFT environment, our results suggest that certain VPIN specifications provide proxies for adverse selection risk similar to those obtained by the PIN model. Thus, we consider that the key variable in the VPIN procedure is the number of buckets used and that VPIN can be a helpful device which is not exclusively applicable to the HFT world.

⑤ Adverse Selection 下的 Protection

(Two Speed Solution / yellow flag)

② 动态调整 matching engine 速度，而不是直接熔断 (circuit break / Red flag)

① 设计基于 Adverse Selection 风险的 futures contract

△ Makers 挂单提供流动性 <=> sell 一个基于 Adverse 的 Implied Option

Option: short a straddle

$$\text{Premium} = \frac{A^* - B^*}{2}$$, A^*, B^* 是 maker 的报价

Taker 需要流动性，支付了这个 Premium

↓ Maker 只能被动的挂单，无选择权

Taker 主动选择权

⇒ 当 Taker 买入：(吃掉 Maker Ask) Taker 行使 Call option / Maker short call

当 Taker 卖出：(吃掉 Maker Bid) Taker 行使 Put option / Maker short put

⇒ Maker: short call + short put

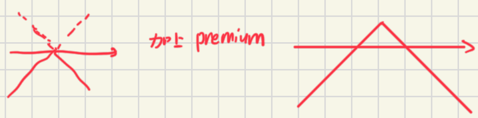
= short straddle

风险：Informed Traders 到来

<=> MM 的 option 被行权

<=> 卖早/买亏 ⇒ 亏损

PNL:



Profit zone: 无单边突破

Loss zone: Informed Trades

$$Profit = \frac{(S_G - S_B)\kappa}{2}(1 - PIN^*)(PIN^* - PIN) .$$

where κ is a constant that relates the volume traded to the range at which liquidity is provided,

$$V = \kappa\left(1 - \frac{A^* - B^*}{S_G - S_B}\right) .$$

e.g.

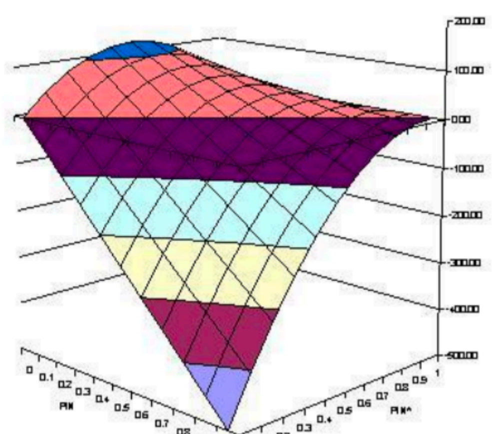
- **场景：** 发生了**知情交易**（Informed Trading）。
  - 比如，一个大户知道某股票马上要从 100 涨到 110。
  - 当前的盘口是：Bid 100 / Ask 100.10。
- **过程：**
  1. 大户（知情者）行权：他以 100.10 的价格买光了 MM 的货（MM Short Call 被行权）。
  2. **MM 的持仓：** 变成了**空头 (Short Position)**，成本价 100.10。
  3. **随后价格走势：** 价格迅速涨到 110。
- **MM 的亏损计算：**

$$PnL = \text{Earned Premium} - \text{Loss from Price Move}$$

$$PnL = (100.10 - 100) - (110 - 100.10) \approx -9.8$$

(为了赚 0.1 的 Spread，亏了 9.9 的价差)

盈亏关键在于对 PIN 的预测误差



The dilemma faced by market makers when proving liquidity under asymmetric information. This surface shows the profit associated with the correct estimation of PIN, and the much greater losses that result from its underestimation

图中的 3D 曲面图展示了 MM 的利润函数。

- **核心变量：** MM 需要猜测当前的 PIN 值（知情交易概率）。设 MM 猜测的值为 PIN^*，实际值为 PIN。
- **残酷的现实：**
  - 如果 MM **猜对了**（$PIN^* \approx PIN$）：他能赚取正常的价差利润（图中平缓的黄色/绿色区域）。
  - 如果 MM **低估了风险**（$PIN^* < PIN$）：即他以为市场很安全，其实知情者正在大量进场。此时利润会**断崖式下跌**（图中深紫色的深坑）。
- **后果：** 因为“低估风险”带来的亏损远远大于“估对风险”带来的利润，MM 会变得**极度保守**。一旦他们察觉到一点点知情交易的迹象（毒性），他们就会立刻清空库存、抽走流动性。这就是 **2010 年闪崩（Flash Crash）** 的微观诱因。

解决方法：

设计 Futures Contract：

Underlying：PIN 数值

Makers：Long　当 PIN ↑，Maker 现货亏钱，期货赚钱

Takers：Short　做为 Informed Trader，他知道自己什么时候 Trade 结束后，short PIN / Toxicity ⇒ 赚钱

⊛ VPIN → execution

VPIN and execution

A large order reveals information to other market participants, who may take advantage of that information by frontrunning that order before its completion. The purpose of an *Optimal Execution Horizon* (OEH) model is to compute the trading horizon that minimizes that informational leakage without incurring in unnecessary market risk. The length of that optimal horizon depends of factors such as: The size of the order, the side, the prevalent order imbalance, market volatility, the trading range and risk aversion. From a theoretical perspective, OEH explains why market participants may rationally 'dump' their orders in an increasingly illiquid market. OEH has been shown to perform better than participation rate schemes. This model is derived as follows:

We have seen earlier that when $\delta_t = \dfrac{1}{2}$, we obtain that

$$\mathrm{E}[A_t - B_t] = \frac{\alpha_t \mu}{\alpha_t \mu + 2\epsilon}(S_G - S_B) \approx \frac{\mathrm{E}[|V^S - V^B|]}{V}(S_G - S_B) \ .$$

This means that we would like to compute V such that $\dfrac{\mathrm{E}[|V^S - V^B|]}{V}$ is minimally impacted. First, ceteris paribus, the impact of an order m on the order imbalance over the next bucket V is

$$\frac{|V^S - V^B|}{V} = \left| \left(2v^B - 1\right)\left(1 - \frac{|m|}{V}\right) + \frac{m}{V} \right| .$$

m : order size

当前 OI

where $v^B = \dfrac{V^B}{V}$ and $\left(2v^B - 1\right)$ is the order imbalance prior to m. Let $\phi\left[|m|\right]$ be a monotonic increasing function of $|m|$, $0 < \phi\left[|m|\right] < 1$, which measures the displacement that $|m|$ causes to the previous order imbalance. Then, the new persistent order imbalance can be computed as

市场记忆力

$$\tilde{OI} = \phi\left[|m|\right]\left[\left(2v^B - 1\right)\left(1 - \frac{|m|}{V}\right) + \frac{m}{V}\right] + \left(1 - \phi\left[|m|\right]\right)\left(2v^B - 1\right) \ .$$

风险厌恶系数

Second, assuming that prices follow an arithmetic random walk, for a risk aversion λ we can derive a timing risk

$$P\left[sgn\left(m\right)\Delta S > Z_\lambda \sigma \left(\frac{V}{V_\sigma}\right)^{\frac{1}{2}}\right] = 1 - \lambda \ .$$

Then, a *probabilistic loss function* can be defined as the aggregation of the first (liquidity risk) and second (timing risk) components,

liquidity Risk : Timing Risk

m↑, 冲击↑

$$\Pi\left(V\right) = \left|\phi\left[|m|\right]\left[\left(2v^B - 1\right)\left(1 - \frac{|m|}{V}\right) + \frac{m}{V}\right] + \left(1 - \phi\left[|m|\right]\right)\left(2v^B - 1\right)\right|\left(S_G - S_B\right) - Z_\lambda \sigma \left(\frac{V}{V_\sigma}\right)^{\frac{1}{2}}$$

Finally, $\Pi\left(V\right)$ can be minimized with respect to V. A novel feature of this model is that, beyond the order size, other important variables are used in determining the optimal V, like the side of the order, and the prevalent order imbalance.

OEH 在两种风险中 trade off:

① V 拆得太快: liquidity Risk / 信息泄漏

② V 拆得太慢: Timing Risk

⊛ VPIN 原文一些研究细节

《 Flow toxicity and liquidity in a High-frequency World 》

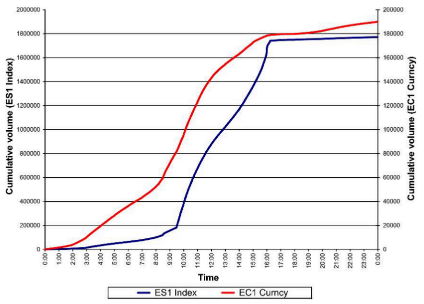


Figure 1
Average daily volume profile of E-mini S&P 500 futures and EC1 futures
This figure shows the intraday volume pattern in the E-mini S&P 500 futures contract and the euro/U.S. dollar futures contract (the EC1 contract) based on a sample from January 1, 2008 to August 15, 2011. E-mini volume is measured in contracts on the left axis, while the EC1 volume is measured in contracts on the right axis.

关于 volume time：
CME 为例：
日盘在 volume 显著大于夜盘

这个图有意思：

Time gap
VPIN
⟺ Time

VPIN 等 volume time，涉及划分 bucket
用 Time gap 可以刻画 算法影响
volume 分布

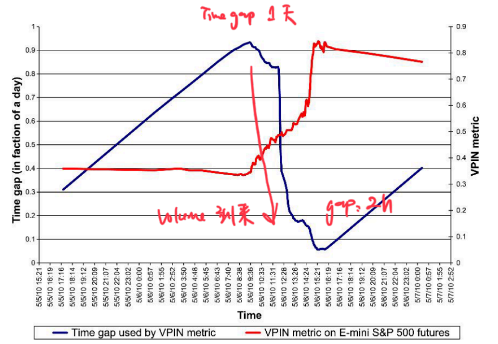


Figure 2
Time gap and VPIN on May 6, 2010
This figure demonstrates how the time gap used to compute the E-mini S&P futures contract changed over the course of May 6. The figure also shows how the VPIN metric changed over the day. The time gap is measured on the left axis, while the VPIN is measured over the right axis.

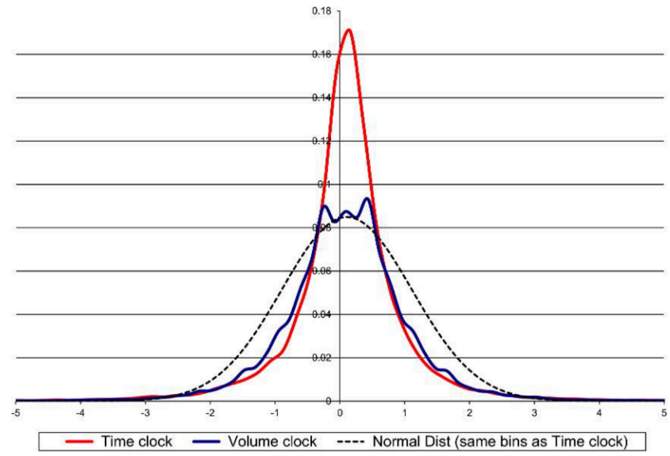


Figure 3
Graphs of price changes for E-mini S&P 500 futures, sampled by regular time intervals and regular volume intervals
This figure shows the distribution of normalized price changes for the E-mini S&P futures contract. Our sample period is January 1, 2008–August 1, 2011. We draw an average of fifty price observations per day, equally spaced in time for the time clock and equally spaced by volume for the volume clock. We compute first-order differences and standardize each sample. The black line gives the normal distribution.

Return 在 volume time 更少出现尖峰厚尾

右图：
特征关于 Computation 中不同算法、参数
稳定性检验

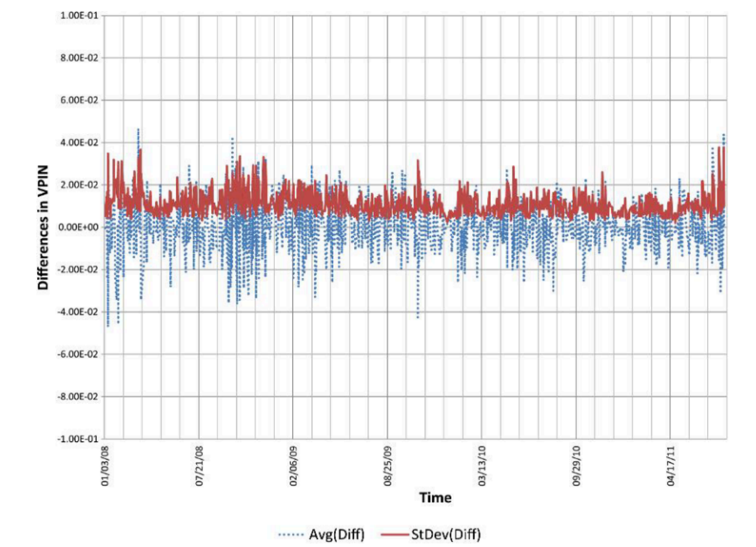


Figure 5
Stability of VPIN toxicity metric estimates
This figure shows the differences in VPIN and their cross-sectional standard deviations from computing 1,000 alternative VPIN trajectories for the E-mini S&P futures contract.

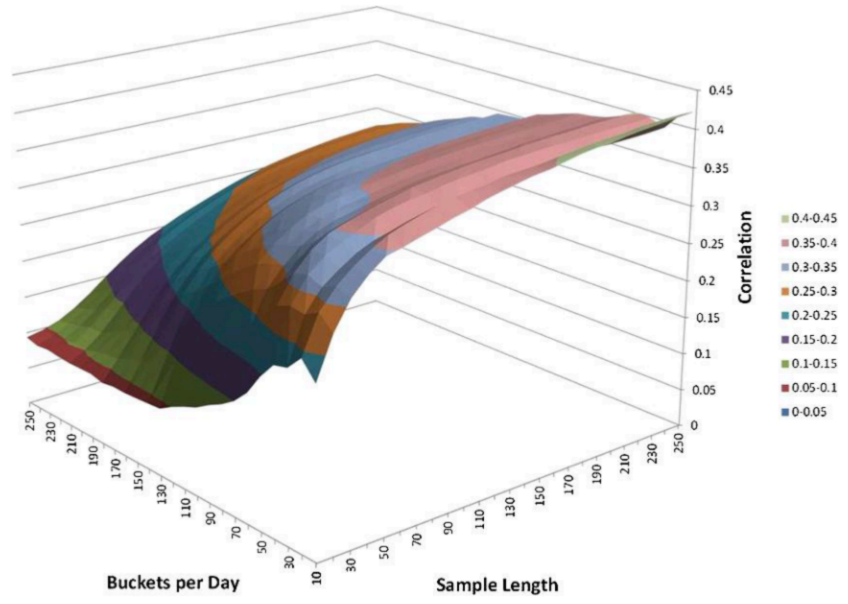


Figure 10
Correlation between VPIN and future volatility for E-mini S&P 500
This figure shows the correlation between VPIN and the following absolute returns for the E-mini S&P futures contract for various combinations of buckets per day and the number of buckets used to calculate VPIN (the sample length).

$\rho(\ln(VPIN), |\frac{R_t}{R_{t-1}} - 1|)$

与未来 absolute returns / volatility

“涉及两个参数调参时”

当 feature 和 $Ret_{t,t+1}$

散点图呈锥形时，

说明和 volatility 呈线性关系

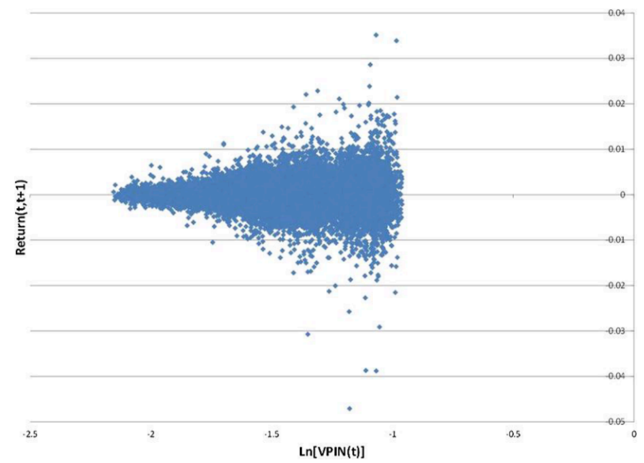


Figure 11
S&P 500's response to VPIN
This figure shows the return on the E-mini S&P 500 futures over the next volume bucket (1/50 of an average day's volume using the (*50,250*) combination) sorted by the log of the previous VPIN level.

Table 4
Conditional probability distributions of VPIN and absolute return

Panel A: Absolute return between two consecutive buckets

VPIN percentiles	0.25%	0.50%	0.75%	1.00%	1.25%	1.50%	1.75%	2.00%	>2.00%
0.05	96.56%	3.33%	0.11%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.10	96.33%	3.44%	0.23%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.15	91.54%	7.56%	0.73%	0.11%	0.06%	0.00%	0.00%	0.00%	0.00%
0.20	90.41%	8.24%	0.96%	0.28%	0.11%	0.00%	0.00%	0.00%	0.00%
0.25	90.74%	8.74%	0.79%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.30	89.54%	9.87%	0.62%	0.00%	0.06%	0.00%	0.00%	0.00%	0.00%
0.35	88.21%	10.55%	1.02%	0.06%	0.17%	0.00%	0.00%	0.00%	0.00%
0.40	84.72%	13.20%	1.69%	0.23%	0.17%	0.00%	0.00%	0.00%	0.00%
0.45	80.88%	17.26%	1.64%	0.17%	0.06%	0.00%	0.00%	0.00%	0.00%
0.50	81.90%	14.89%	2.65%	0.34%	0.06%	0.17%	0.00%	0.00%	0.00%
0.55	79.74%	17.55%	2.09%	0.56%	0.00%	0.06%	0.00%	0.00%	0.00%
0.60	79.30%	18.05%	2.09%	0.39%	0.11%	0.00%	0.06%	0.00%	0.00%
0.65	75.06%	16.92%	2.88%	0.39%	0.28%	0.11%	0.06%	0.00%	0.00%
0.70	68.25%	20.60%	3.22%	0.85%	0.11%	0.06%	0.06%	0.06%	0.00%
0.75	62.32%	24.99%	5.25%	1.18%	0.11%	0.11%	0.11%	0.00%	0.00%
0.80	62.81%	27.58%	6.49%	2.65%	0.51%	0.28%	0.17%	0.00%	0.00%
0.85	56.38%	26.52%	7.67%	1.86%	0.51%	0.23%	0.34%	0.06%	0.00%
0.90	43.71%	29.35%	9.20%	2.93%	1.24%	0.51%	0.06%	0.11%	0.23%
0.95	39.56%	30.51%	16.02%	5.75%	2.14%	0.79%	0.51%	0.17%	0.39%
1.00	39.56%	29.12%	16.42%	7.62%	3.27%	1.64%	0.90%	0.73%	0.73%

Table 4a $-Prob\left(\left|\frac{P_\tau}{P_{\tau-1}} - 1\right| \middle| VPIN_{\tau-1}\right)$

Panel B: Absolute return between two consecutive buckets

VPIN percentiles	0.25%	0.50%	0.75	1.00%	1.25%	1.50	2.00%	>= 2%
0.05	6.28%	0.98%	0.14%	0.00%	0.00%	0.00%	0.00%	0.00%
0.10	6.27%	1.02%	0.28%	0.00%	0.00%	0.00%	0.00%	0.00%
0.15	5.96%	2.23%	0.90%	0.44%	0.63%	0.00%	0.00%	0.00%
0.20	5.88%	2.43%	1.17%	1.11%	1.26%	0.00%	0.00%	0.00%
0.25	5.89%	2.59%	0.97%	0.00%	0.00%	0.00%	0.00%	0.00%
0.30	5.82%	2.92%	0.76%	0.00%	0.63%	0.00%	0.00%	0.00%
0.35	5.74%	3.12%	1.24%	0.22%	1.89%	0.00%	0.00%	0.00%
0.40	5.51%	3.90%	2.07%	0.89%	1.89%	0.00%	0.00%	0.00%
0.45	5.26%	5.10%	2.00%	0.67%	0.63%	0.00%	0.00%	0.00%
0.50	5.33%	4.40%	3.24%	1.33%	0.63%	4.29%	0.00%	0.00%
0.55	5.19%	5.19%	2.55%	2.22%	0.00%	1.43%	0.00%	0.00%
0.60	5.16%	5.34%	2.55%	1.56%	1.26%	0.00%	2.50%	0.00%
0.65	5.16%	5.00%	3.52%	1.56%	3.14%	2.86%	2.50%	2.22%
0.70	4.88%	6.09%	3.93%	3.33%	1.26%	1.43%	2.50%	2.22%
0.75	4.44%	7.39	6.42%	4.67%	1.26%	2.86%	5.00%	0.00%
0.80	4.06%	8.16%	7.94%	10.44%	5.66%	7.14%	7.50%	0.00%
0.85	4.09%	7.84%	9.39%	7.33%	5.66%	5.71%	15.00%	2.22%
0.90	3.67%	8.67	11.25%	11.56%	13.84%	12.86%	2.50%	13.33%
0.95	2.84%	9.02%	19.60%	22.67%	23.90%	20.00%	22.50%	22.22%
1.00	2.57%	8.61%	20.08%	30.00%	36.48%	41.43%	40.00%	57.78%

Table 4b $- Prob\left(VPIN_{\tau-1} \middle| \left|\frac{P_\tau}{P_{\tau-1}} - 1\right|\right)$

This table gives the conditional probability distributions of the VPIN toxicity metric and absolute return. Table 4(a) shows distributions of returns at time τ given VPIN at time $\tau-1$. Table 4(b) shows distributions of VPIN at time $\tau-1$ given returns at time τ.

Joint Distribution 和 Conditional Distribution

$VPIN_{t-1}$ 和 $|Ret_{t+1}|$

分别控制 conditions 水平，

去观察变量分布

⇒ 对观察到的一些头尾极端预测能力

后续可单独研究

Volatility，VPIN 关系？

作者设计实验： VPIN 和 volatility 异质

以及： VPIN high 一定预测 volatility ↑ ？

我们 assume 的是： VPIN 作为 toxic 的指标

它能预测未来极端 vol（比如 crashes）

但不会和未来和 toxic 无关的 vol（比如小波动）相关

To examine the maximum intermediate volatility experienced by a market maker while VPIN is high, we compute volatility events while VPIN remains within any fifth percentile. In particular, every time VPIN moves from one fifth percentile to another, we compute the largest absolute return that occurs between any two intermediate (not necessarily consecutive) buckets, until VPIN moves to another fifth percentile. For example, suppose that the VPIN metric moves into the eighty-fifth percentile, and four volume buckets later it moves to the ninetieth percentile. We would calculate the absolute return between buckets 1 and 2, 1 and 3, and 1 and 4, as well as between 2 and 3, 2 and 4, and 3 and 4. We are interested in the maximum of these absolute returns, which we view as the maximum volatility the market maker faces. We do this for every such VPIN crossing.

To be precise, we let i be an index that is updated by one every time that VPIN crosses from one percentile into another, and $\tau(i)$ the bucket associated with the ith cross. This means that VPIN remained in the same percentile for $\tau(i+1) - \tau(i)$ buckets; in our example above, this was the four-bucket interval over which VPIN remained in the eighty-fifth percentile. The largest absolute return between any two intermediate buckets while VPIN remained in a particular percentile (i.e., from the bucket at which it made the ith crossing until it made the $i + 1$th crossing) is then

$$\max_{\substack{\tau(i) \le m < l \\ \tau(i) < l \le \tau(i+1)}} \left| \frac{P_l}{P_m} - 1 \right|. \tag{10}$$

In our example above, this is the maximum of the six intermediate returns over the interval in which VPIN remained in the eighty-fifth percentile.

This analysis is richer than the conditional probabilities illustrated in Table 4a in two respects. First, we are not just considering the absolute return that occurs in the bucket immediately following VPIN's crossing into a bin, but rather the maximum absolute return that occurs while VPIN remains in a bin. This is consistent with our microstructure theory that volatility appears once toxicity has reached a saturation point that exceeds market makers' tolerance. Second, it captures price volatility across all sequences of intermediate buckets, thus incorporating the effect of price drifts as well as price recoveries. This is important, because slow price drifts and price recoveries may both hide extreme volatility. But this analysis is also limited in that it does not capture sustained increases in toxicity spanning multiple VPIN percentiles. In particular, we do not capture what happens when toxicity increases from the seventy-fifth to the eightieth and then on to the eighty-fifth, ninetieth, etc. Thus, the hurdle we set here will surely underestimate the effect of toxicity-induced volatility.

Figure 14 plots the probabilities of the largest absolute return being in excess of 0.75% while VPIN remains in any fifth percentile in the upper quartile of its distribution (i.e., VPIN is in the 0.75–0.80, 0.80–0.85, 0.85–0.90, 0.90–0.95, or 0.95–1.00 bin) for various combinations of buckets per day and sample length. For our standard combination of *(50, 250)*, we find that 51.84% of the times that VPIN enters a fifth percentile within the upper quartile there is at least one intermediate return in excess of 0.75% before VPIN leaves that fifth percentile. The parameter combination that maximizes VPIN's predictive power is *(10, 350)*—that is, ten buckets per day for a sample length of 350 (about 1.6 months). For this parameter combination, Table 5 shows that 78.57% of the times that VPIN enters a fifth percentile within the upper quartile of its distribution, toxicity-induced volatility often will be substantial (absolute returns in excess of 0.75%) before VPIN leaves that fifth percentile.

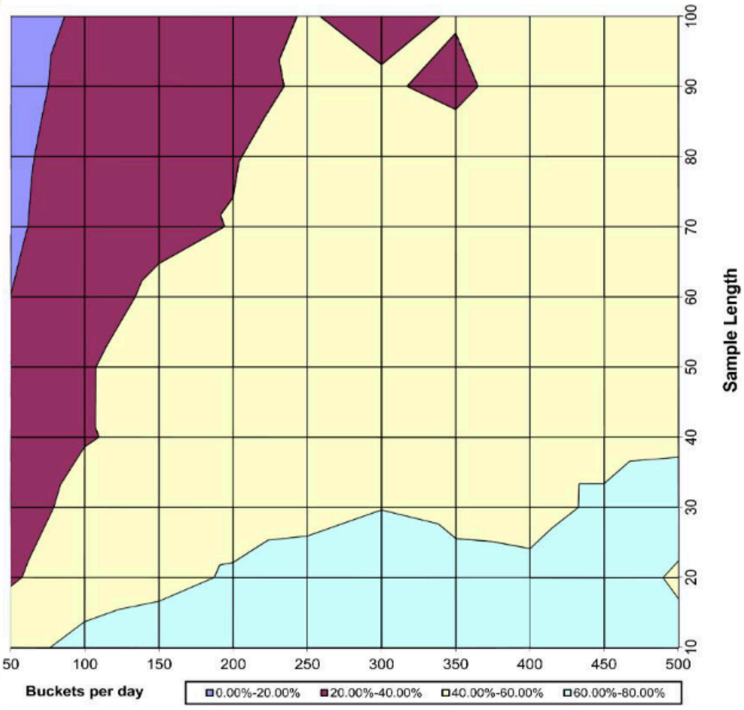


Figure 14
Probability of "true positives"
This figure plots the probabilities of the largest absolute return being in excess of 0.75% while VPIN remains in any fifth percentile in the upper quartile of its distribution for various combinations of buckets per day and sample length.

Table 5
VPIN and volatility events

Absolute return between any two intermediate buckets

VPIN percentiles	0.25%	0.50%	0.75%	1.00%	1.25%	1.50%	1.75%	2.00%	>2.00%
0.05	25.00%	16.67%	25.00%	8.33%	0.00%	8.33%	8.33%	8.33%	0.00%
0.10	27.27%	27.27%	3.03%	9.09%	3.03%	0.00%	9.09%	0.00%	21.21%
0.15	39.58%	12.50%	10.42%	4.17%	6.25%	4.17%	2.08%	8.33%	12.50%
0.20	30.99%	25.35%	15.49%	8.45%	9.86%	0.00%	4.23%	1.41%	4.23%
0.25	21.69%	19.28%	22.89%	12.05%	6.02%	9.64%	2.41%	2.41%	3.61%
0.30	17.50%	25.00%	20.00%	13.75%	7.50%	3.75%	3.75%	2.50%	6.25%
0.35	19.05%	26.19%	16.67%	13.10%	5.95%	8.33%	5.95%	2.38%	2.38%
0.40	10.71%	16.67%	23.81%	13.10%	11.90%	7.14%	4.76%	2.38%	9.52%
0.45	15.91%	19.32%	13.64%	13.64%	14.77%	9.09%	6.82%	1.14%	5.68%
0.50	19.59%	16.49%	19.59%	16.49%	6.19%	4.12%	5.15%	4.12%	8.25%
0.55	18.39%	13.79%	18.39%	13.79%	11.49%	4.60%	6.90%	4.60%	8.05%
0.60	14.04%	10.53%	21.05%	17.54%	7.02%	7.02%	1.75%	10.53%	10.53%
0.65	12.00%	12.00%	8.00%	10.00%	14.00%	8.00%	2.00%	12.00%	22.00%
0.70	12.07%	10.34%	12.07%	8.62%	6.90%	8.62%	6.90%	6.90%	27.59%
0.75	14.04%	14.04%	10.53%	15.79%	5.26%	3.51%	10.53%	0.00%	26.32%
0.80	10.53%	8.77%	7.02%	8.77%	10.53%	3.51%	1.75%	7.02%	42.11%
0.85	8.33%	13.89%	5.56%	8.33%	11.11%	8.33%	5.56%	5.56%	33.33%
0.90	0.00%	0.00%	0.00%	10.00%	10.00%	20.00%	0.00%	10.00%	50.00%
0.95	0.00%	7.69%	7.69%	7.69%	0.00%	0.00%	0.00%	0.00%	76.92%
1.00	0.00%	0.00%	0.00%	0.00%	10.00%	0.00%	0.00%	0.00%	90.00%

This table shows the maximum exposure of the market maker to price movements following a transition of the VPIN metric from one level to the next. It is computed using ten buckets per day and a sample length of 350 (about 1.6 months).